

CPU Scheduling Comparison Report

This report presents a professional comparison between different CPU Scheduling Algorithms including SJF, SRTF, Priority Preemptive, and Priority Non-Preemptive. The selected test cases are designed to demonstrate the fundamental differences between preemptive and non-preemptive scheduling techniques.

Test Case 1 — Non-Preemptive Scheduling Comparison

Process	Arrival Time	Burst Time	Priority
P1	0	5	2
P2	1	3	1
P3	2	1	3

Purpose:

This test case demonstrates the behavior of non-preemptive scheduling algorithms. It shows how a running process continues execution until completion even if a shorter or higher-priority process arrives later.

SJF Non-Preemptive Gantt Chart

```
0  -----5---6-----9
|   P1   | P3 |   P2   |
```

Analysis

- P1 started first because it arrived at time 0.
- Since the algorithm is non-preemptive, P1 continued execution until completion.
- After P1 finished, the scheduler selected the shortest burst time process.
- P3 executed before P2 because its burst time was smaller.

What does this show?

- SJF focuses on minimizing burst time.
- Shorter processes are preferred.
- Average waiting time becomes lower.

Priority Non-Preemptive

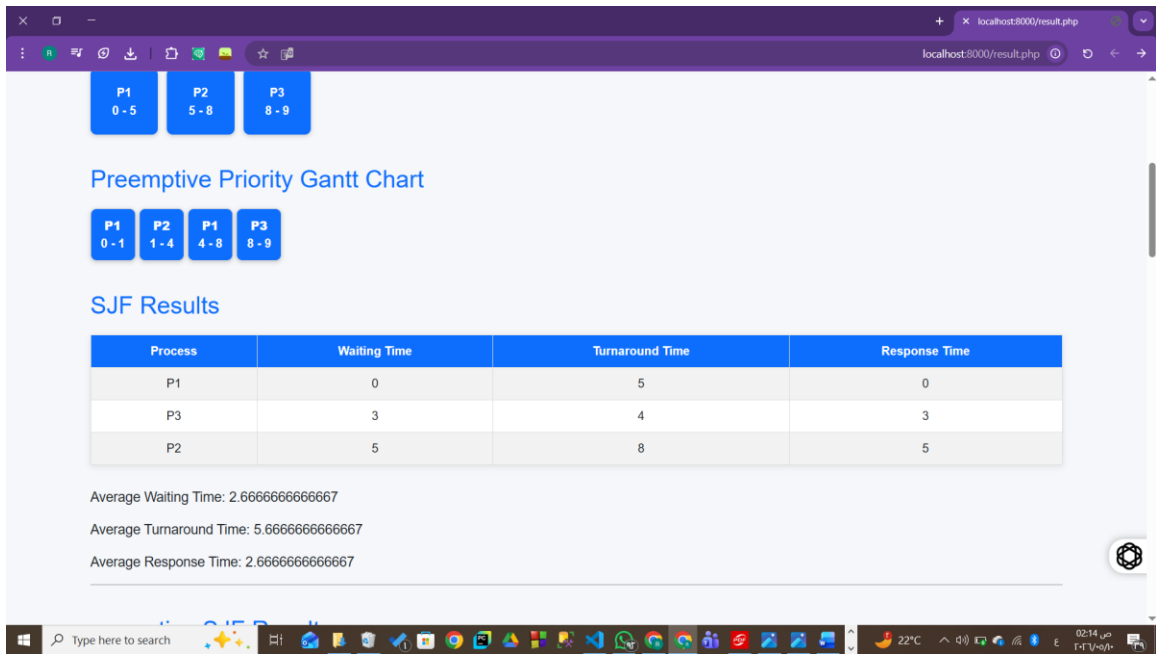
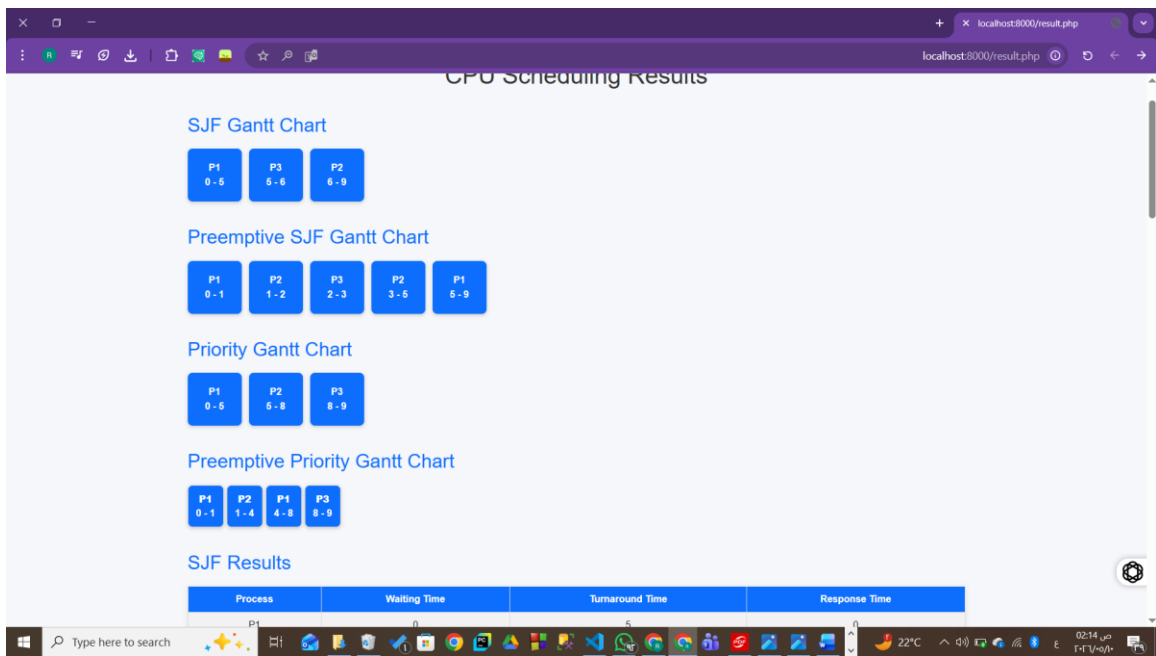
```
0  -----5---8-----9
|   P1   | P2 |   P3   |
```

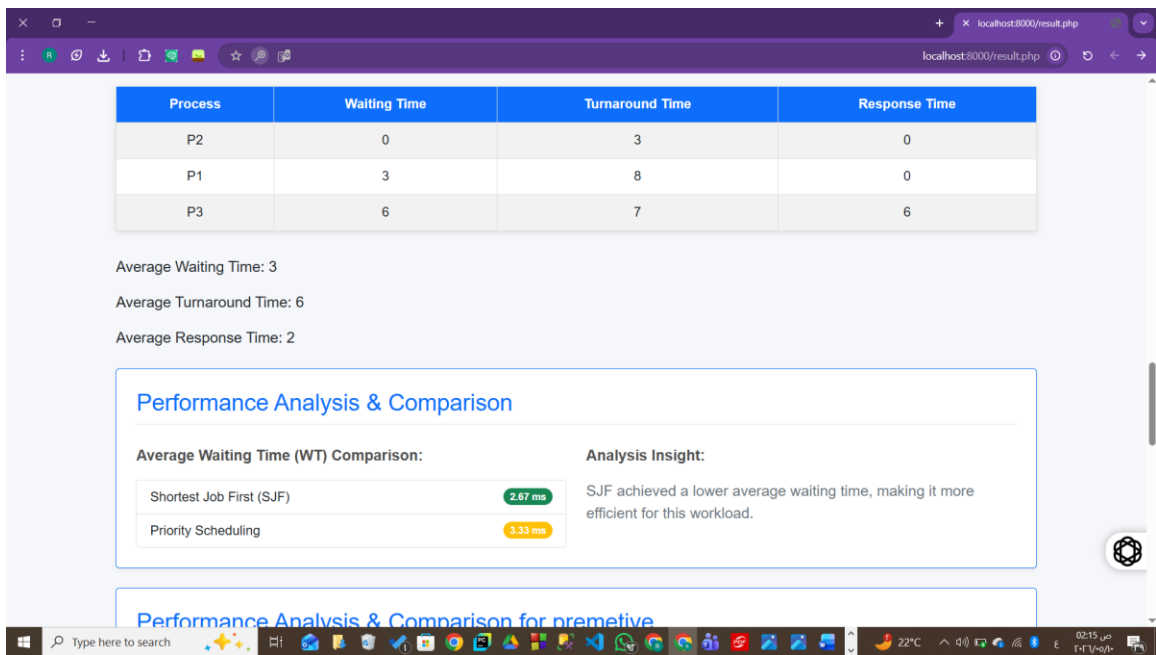
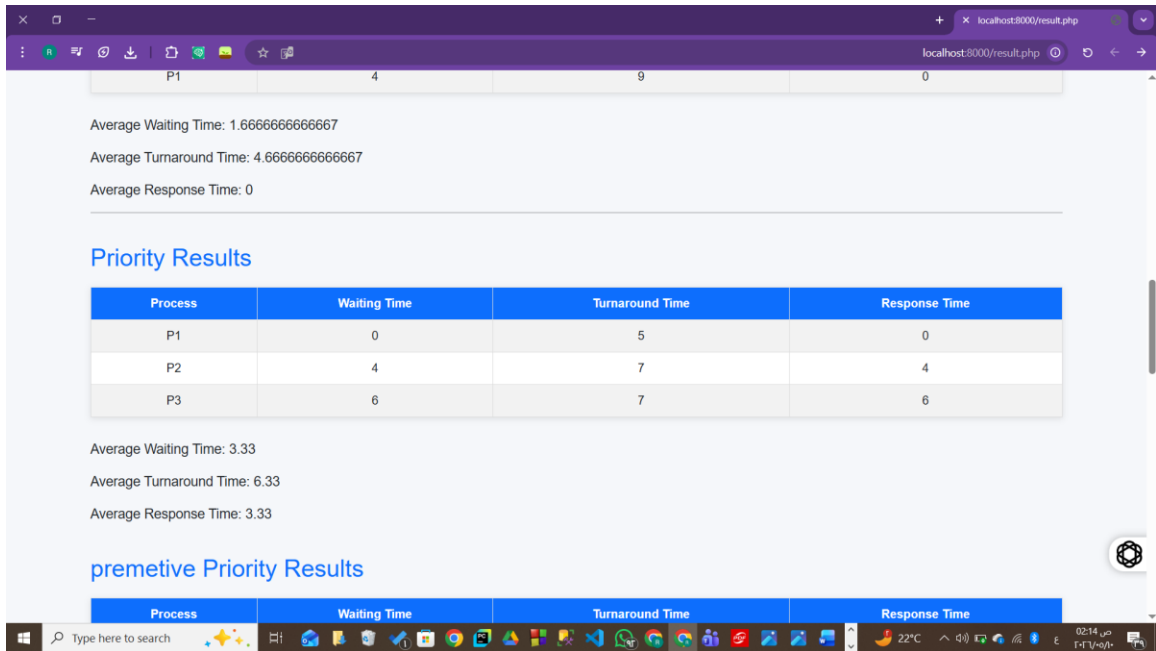
Priority Non-Preemptive

- P1 started execution first and was not interrupted.
- After P1 c
- completed, the scheduler selected the highest priority process.
- P2 executed before P3 because it had higher priority.
- **What does this show?**
- Priority scheduling focuses on process importance.
- Burst time is ignored during selection.
- Important tasks receive CPU access first.
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Process ID	Arrival Time	Burst Time	Priority	Action
P1	0	5	2	X
P2	1	3	1	X
P3	2	1	3	X

+ ADD PROCESS SUBMIT





Test Case 2 — Preemptive Scheduling Comparison

Process	Arrival Time	Burst Time	Priority
P1	0	8	2
P2	1	4	1
P3	2	2	3
P4	3	1	4

Preemptive SJF (SRTF) Gantt Chart

0-1 P1 | 1-2 P2 | 2-4 P3 | 4-5 P4 | 5-8 P2 | 8-15 P1

Explanation:

SRTF always selects the process with the shortest remaining burst time. This improves response time and CPU efficiency but increases context switching.

Priority Preemptive Gantt Chart

0-1 P1 | 1-5 P2 | 5-13 P1 | 13-15 P3 | 15-16 P4

Explanation:

Priority scheduling selects the process with the highest priority (lowest numeric value).

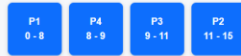
High-priority processes can interrupt currently running processes.

The screenshot shows a web browser window displaying a "CPU Scheduling Simulator" application. The browser's address bar shows "localhost:8000/input.php". The application has a navigation bar with "input" and "result" tabs. The main content area features a form titled "CPU Scheduling Simulator" with a table for inputting process details. The table has columns for Process ID, Arrival Time, Burst Time, Priority, and Action. It contains four rows of data for processes P1, P2, P3, and P4. Below the table are two buttons: "+ ADD PROCESS" and "SUBMIT". The Windows taskbar is visible at the bottom of the screen.

Process ID	Arrival Time	Burst Time	Priority	Action
P1	0	8	2	X
P2	1	4	1	X
P3	2	2	3	X
P4	3	1	4	X

CPU Scheduling Results

SJF Gantt Chart



Preemptive SJF Gantt Chart



Priority Gantt Chart



Preemptive Priority Gantt Chart



Average Waiting Time: 5.5

Average Turnaround Time: 9.25

Average Response Time: 5.5

premetive SJF Results

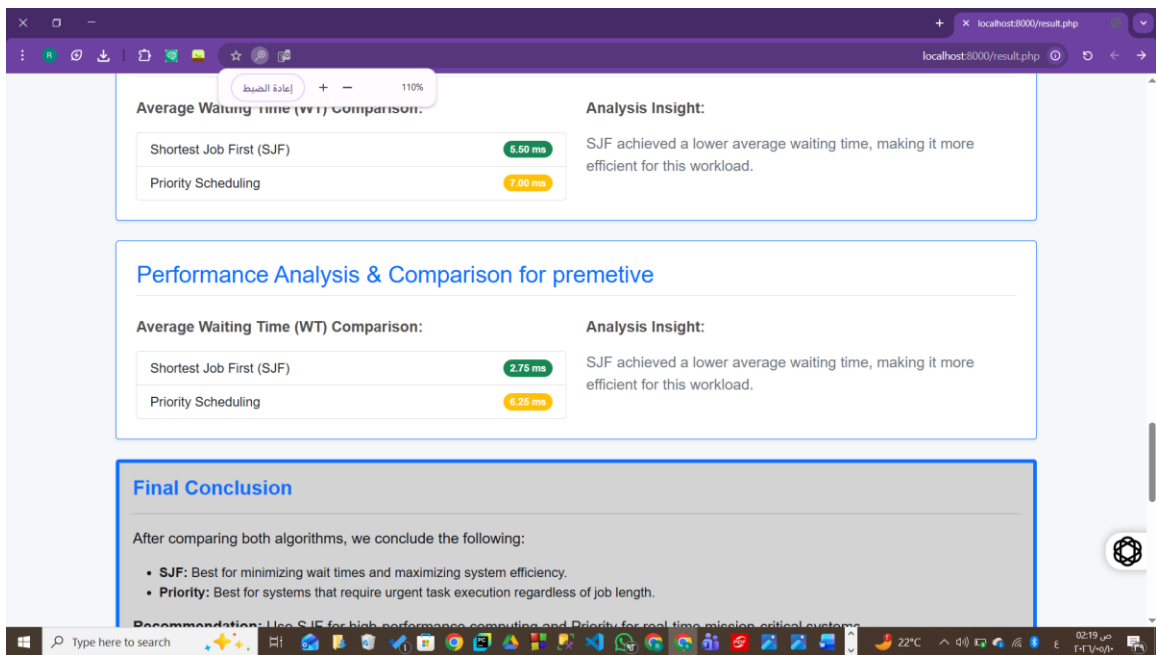
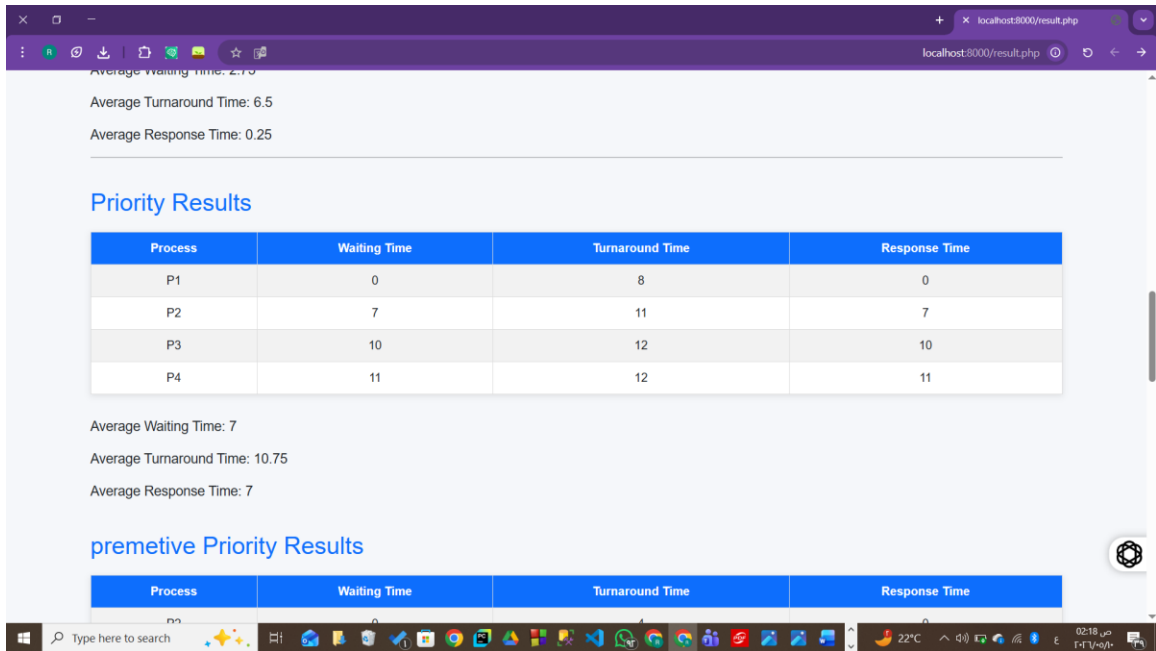
Process	Waiting Time	Turnaround Time	Response Time
P3	0	2	0
P4	1	2	1
P2	3	7	0
P1	7	15	0

Average Waiting Time: 2.75

Average Turnaround Time: 6.5

Average Response Time: 0.25

Priority Results



Test Case 3 — Strong Difference Demonstration

Process	Arrival Time	Burst Time	Priority
P1	0	10	3
P2	1	2	1
P3	2	1	4
P4	3	5	2

P5	4	3	1
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Purpose:

This test case was specifically selected to generate different execution orders for every algorithm. It clearly demonstrates the impact of arrival time, priority, and burst time.

SJF Non-Preemptive

0-----10-11--13---16-----21
| P1 | P3|P2|P5| P4 |

SRTF

0 1 2 3 5 8 13 22
P1 P2 P3 P2 P5 P4 P1

Priority Non-Preemptive

0-----10--12---15-----20--21
| P1 | P2 | P5 | P4 | P3 |

Priority Preemptive

0 1 3 6 11 20 21
P1 P2 P5 P4 P1 P3

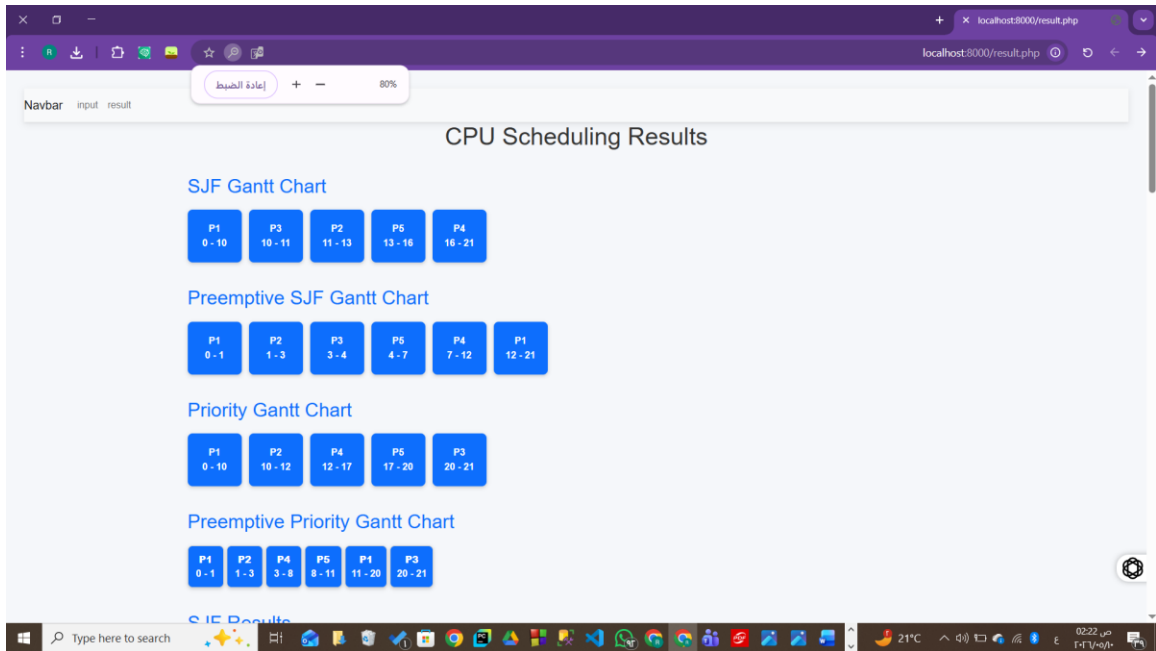
Analysis:

This scenario highlights the core difference between preemptive and non-preemptive scheduling. Preemptive algorithms provide faster response for short or important processes, while non-preemptive algorithms reduce context switching overhead.

The screenshot shows a web browser window with the address bar displaying 'localhost:8000/input.php'. The page has a 'Navbar' with 'input' and 'result' tabs. The main content is a 'CPU Scheduling Simulator' form. The form contains a table with the following data:

Process ID	Arrival Time	Burst Time	Priority	Action
P1	0	10	2	X
P2	1	2	1	X
P3	2	1	4	X
P4	3	5	1	X
P5	4	3	1	X

Below the table are two buttons: '+ ADD PROCESS' (green) and 'SUBMIT' (blue). The browser's taskbar at the bottom shows the time as 02:22 and the date as 14/11/2023.



localhost:8000/result.php

SJF Results

Process	Waiting Time	Turnaround Time	Response Time
P1	0	10	0
P3	8	9	8
P2	10	12	10
P5	9	12	9
P4	13	18	13

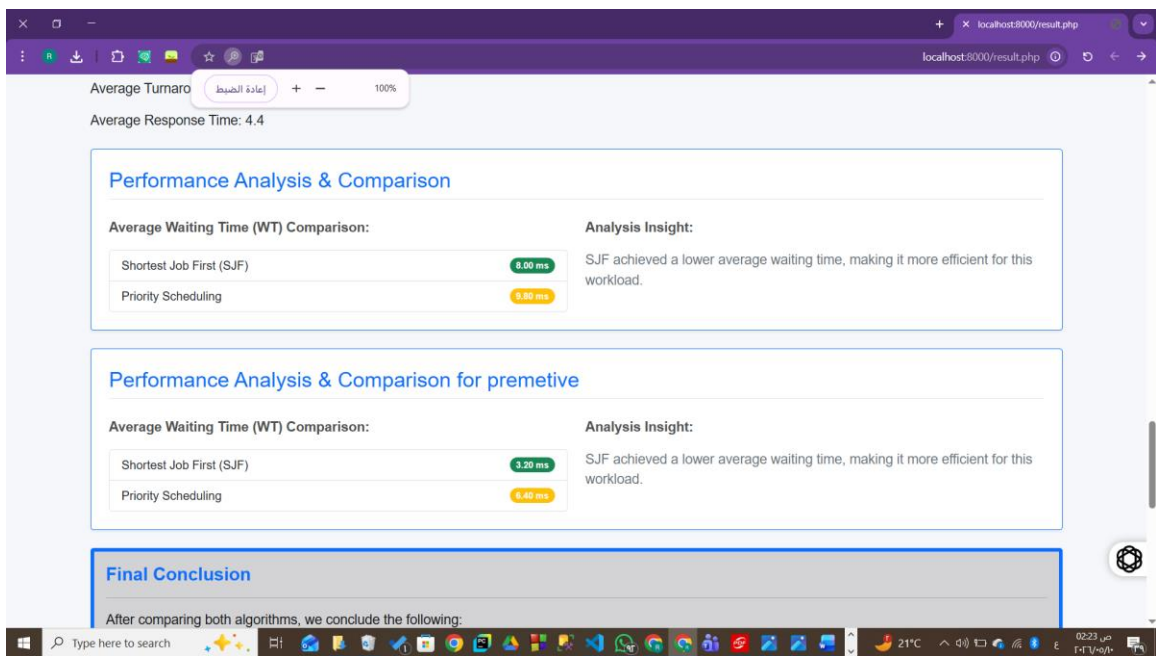
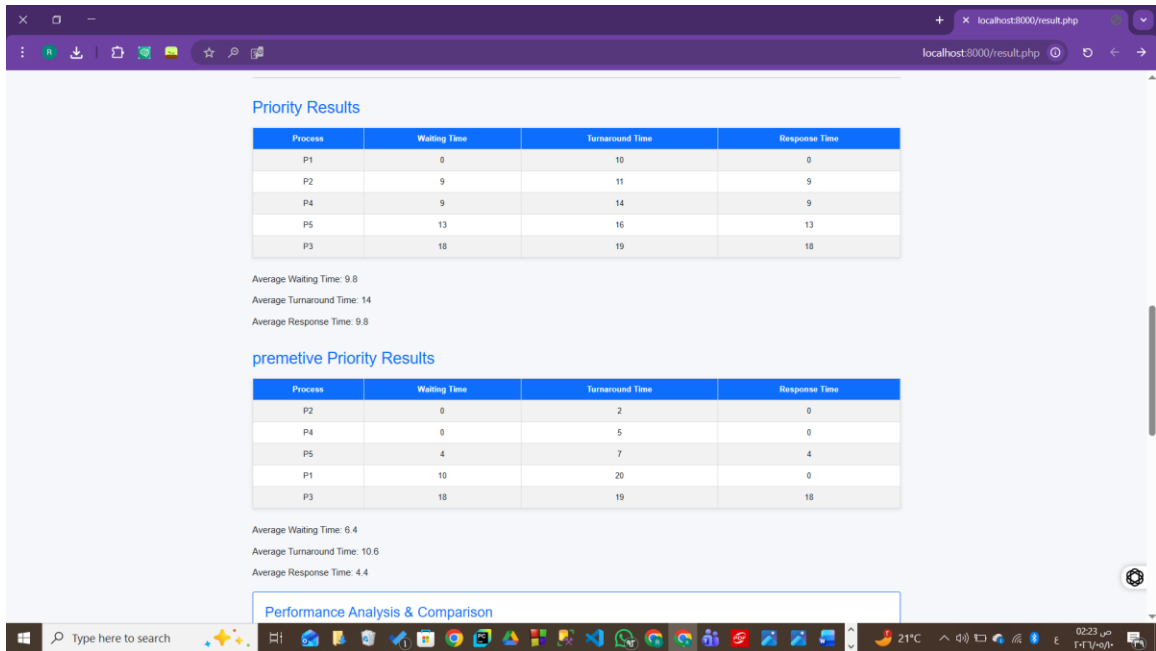
Average Waiting Time: 8
Average Turnaround Time: 12.2
Average Response Time: 8

premetive SJF Results

Process	Waiting Time	Turnaround Time	Response Time
P2	0	2	0
P3	1	2	1
P5	0	3	0
P4	4	9	4
P1	11	21	0

Average Waiting Time: 3.2
Average Turnaround Time: 7.4
Average Response Time: 1

Priority Results



Test Case 4 — Priority Starvation Scenario

Process	Arrival Time	Burst Time	Priority
P1	0	12	5
P2	1	2	1

P3	2	2	1
P4	3	1	1

Purpose:

This test case demonstrates starvation in priority scheduling. Low-priority processes may experience long waiting times if high-priority processes continue arriving.

Priority Preemptive

0-1 P1 | 1-3 P2 | 3-5 P3 | 5-6 P4 | 6-17 P1

Analysis:

The low-priority process P1 was interrupted multiple times. This demonstrates starvation and fairness problems in priority scheduling.

1. SJF (Non-Preemptive)

Gantt Chart

0 ----- 12 -- 14 -- 16 -- 17
| P1 | P4 | P2 | P3 |

Analysis

SJF ignores priority and always selects the process with the smallest burst time available. P1 executes first and completes without interruption, even though higher priority processes arrive later.

2. SRTF (Preemptive SJF)

Gantt Chart

0 1 3 5 6 ----- 17
|P1|P2|P3|P4| P1 |

Analysis

SRTF always selects the process with the shortest remaining time. P1 is repeatedly preempted whenever a shorter job arrives, regardless of priority.

3. Priority Non-Preemptive

Gantt Chart

0 ----- 12 -- 14 -- 16 -- 17
| P1 | P2 | P3 | P4 |

Analysis

The CPU selects processes based on priority, but once a process starts execution, it cannot be interrupted.

P1 runs first until completion, while other higher priority processes wait in the ready queue

The screenshot displays a web application titled "CPU Scheduling Results" and "CPU Scheduling Simulator". The top section shows the results for four scheduling algorithms, each with a Gantt chart visualization. The bottom section is the input form for the simulator.

CPU Scheduling Results

SJF Gantt Chart

Process	Start Time	End Time
P1	0	12
P4	12	13
P2	13	15
P3	15	17

Preemptive SJF Gantt Chart

Process	Start Time	End Time
P1	0	1
P2	1	3
P4	3	4
P3	4	6
P1	6	17

Priority Gantt Chart

Process	Start Time	End Time
P1	0	12
P2	12	14
P3	14	16
P4	16	17

Preemptive Priority Gantt Chart

Process	Start Time	End Time
P1	0	1
P2	1	3
P3	3	5
P4	5	6
P1	6	17

CPU Scheduling Simulator

Process ID	Arrival Time	Burst Time	Priority	Action
P1	0	12	5	X
P2	1	2	1	X
P3	2	2	1	X
P4	3	1	1	X

+ ADD PROCESS **SUBMIT**

localhost:8000/result.php

localhost:8000/result.php

SJF Results

Process	Waiting Time	Turnaround Time	Response Time
P1	0	12	0
P4	9	10	9
P2	12	14	12
P3	13	15	13

Average Waiting Time: 8.5
Average Turnaround Time: 12.75
Average Response Time: 8.5

premetive SJF Results

Process	Waiting Time	Turnaround Time	Response Time
P2	0	2	0
P4	0	1	0
P3	2	4	2
P1	5	17	0

Average Waiting Time: 1.75
Average Turnaround Time: 6
Average Response Time: 0.5

Priority Results

Type here to search

21°C

02:24

localhost:8000/result.php

localhost:8000/result.php

Priority Results

Process	Waiting Time	Turnaround Time	Response Time
P1	0	12	0
P2	11	13	11
P3	12	14	12
P4	13	14	13

Average Waiting Time: 9
Average Turnaround Time: 13.25
Average Response Time: 9

premetive Priority Results

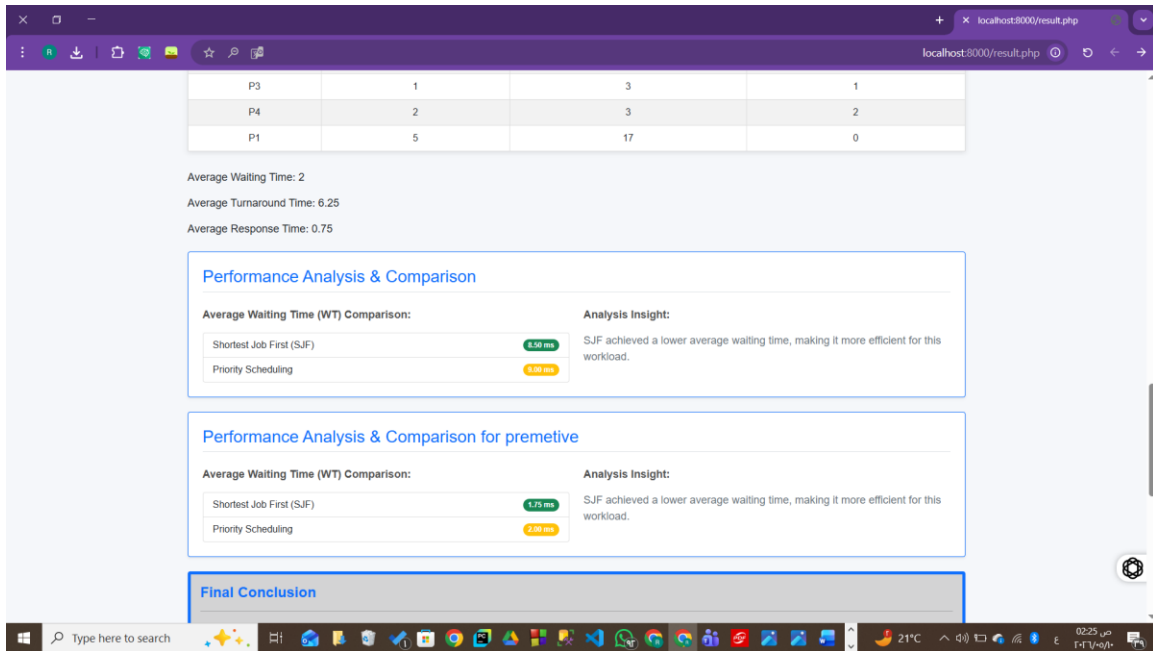
Process	Waiting Time	Turnaround Time	Response Time
P2	0	2	0
P3	1	3	1
P4	2	3	2
P1	5	17	0

Average Waiting Time: 2
Average Turnaround Time: 6.25
Average Response Time: 0.75

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Overall Comparison

Algorithm	Best Feature	Weakness	Recommended Usage
SJF	Low Waiting Time	Poor responsiveness	Batch Systems
SRTF	Fast Response Time	High Context Switching	Interactive Systems
Priority NP	Simple Priority Handling	May delay urgent tasks	Administrative Systems
Priority P	Urgent Task Execution	Starvation Risk	Real-Time Systems

Conclusion

There is no single best CPU scheduling algorithm for all systems. Each algorithm is designed to optimize different performance metrics. SJF and SRTF focus on minimizing waiting and response time, while Priority Scheduling focuses on task importance and urgency.